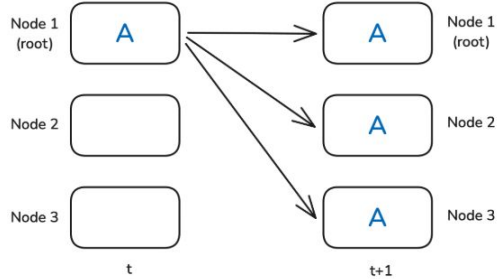


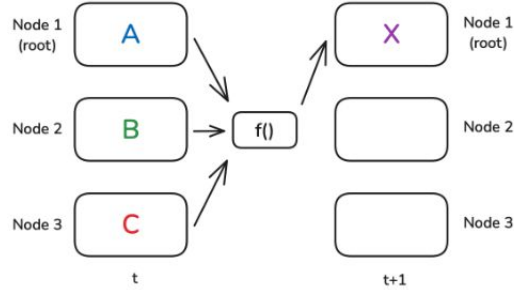
DATA PARALLELISM SLIDES

COLLECTIVE COMMUNICATION PATTERNS

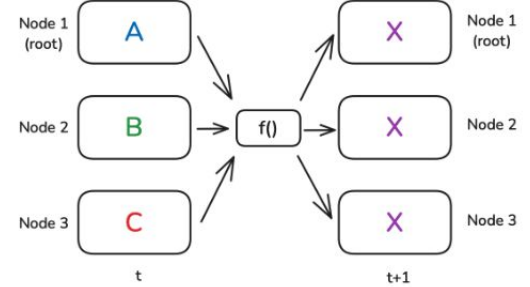
Broadcast



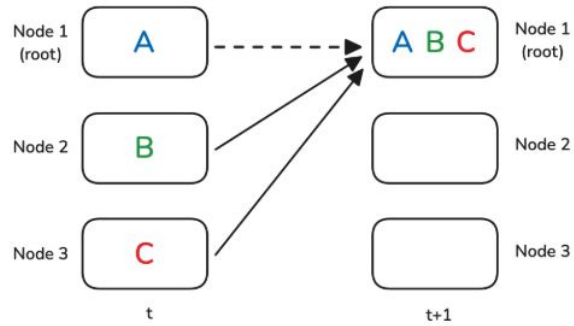
Reduce



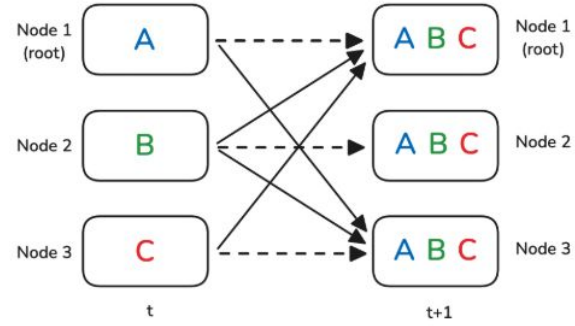
AllReduce



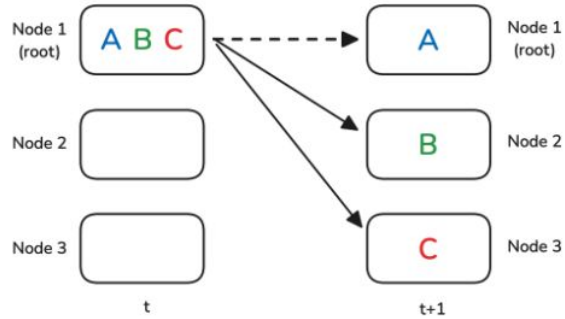
Gather



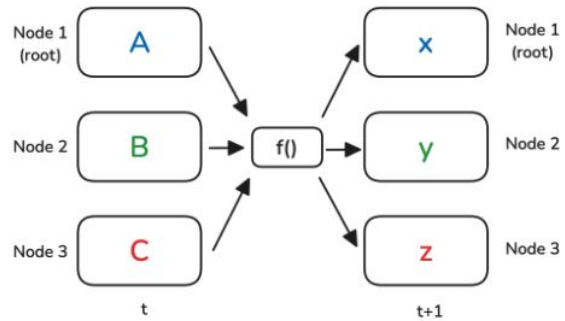
AllGather

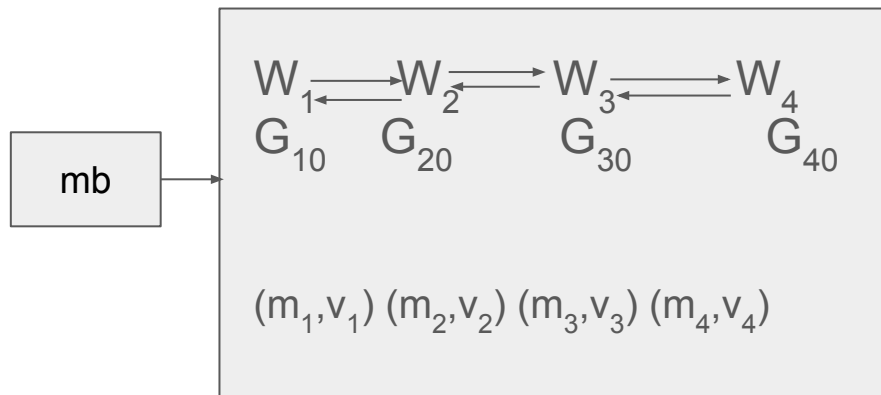


Scatter



ReduceScatter





W_i Parameter

G_{ij} Gradient for i and device j

m_i first moment for parameter i

v_i second moment for parameter i

Memory (N parameters)

Weight = $2 * N$ (bf16)

Gradients = $4 * N$ (fp32)

Optimizers = $4 * N + 4 * N$ (fp32)

Master weights = $4 * N$ (fp32)

Update parameters (W1)

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

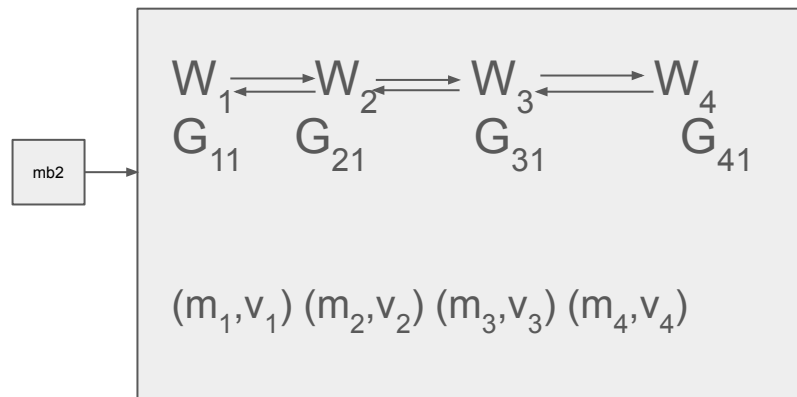
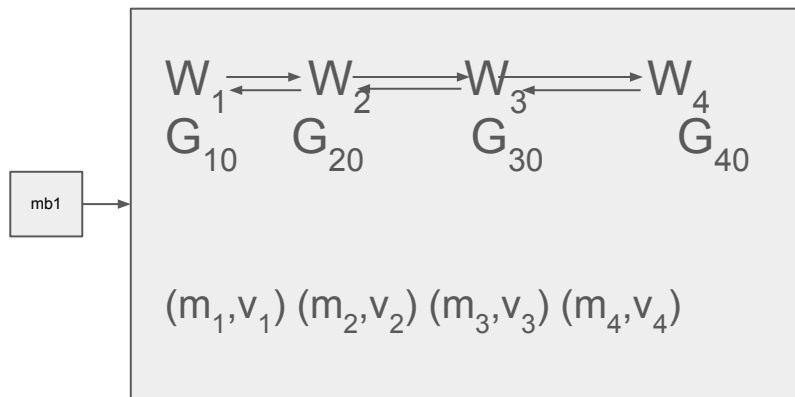
$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$$

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t}$$

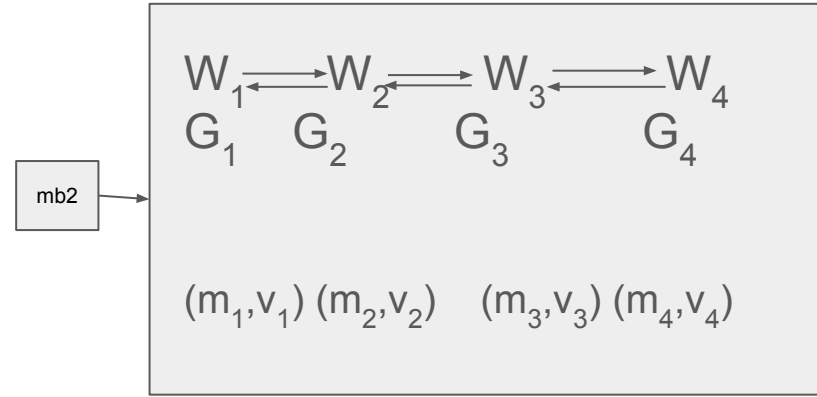
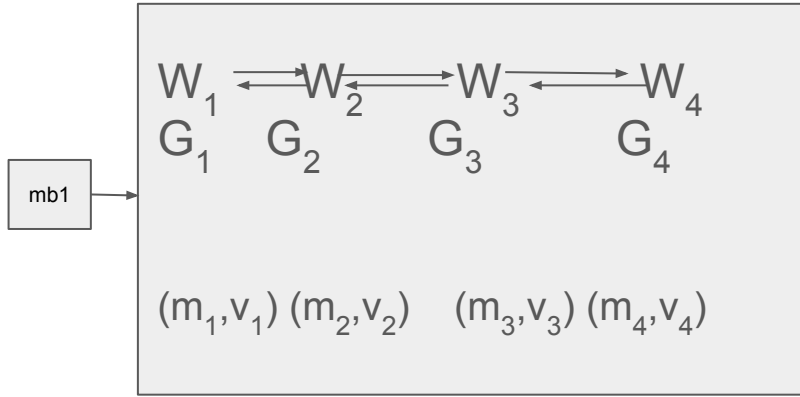
$$\theta'_t = \theta_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$$

Data Parallel



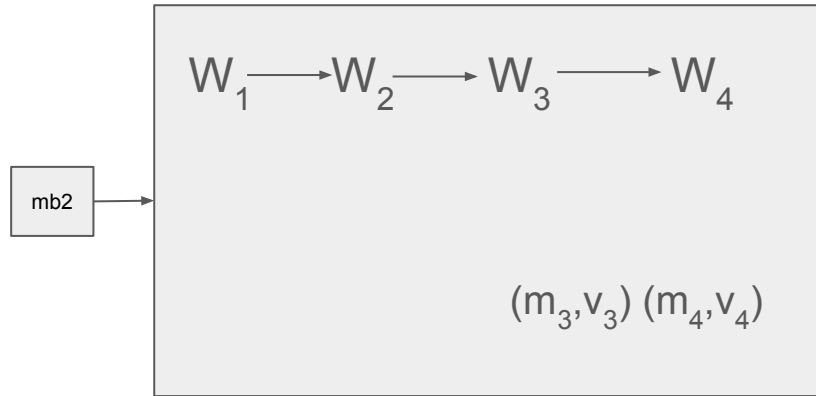
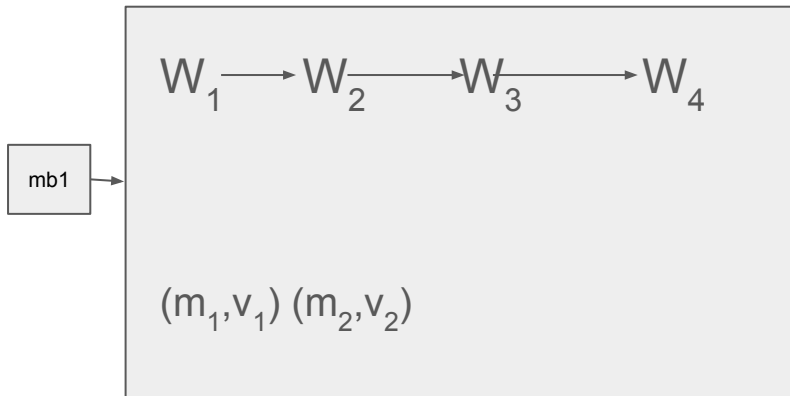
Communication (world size = 2)

- All-reduce the gradients
 - $G1 = (G10 + G11)/\text{world_size}$ and so on

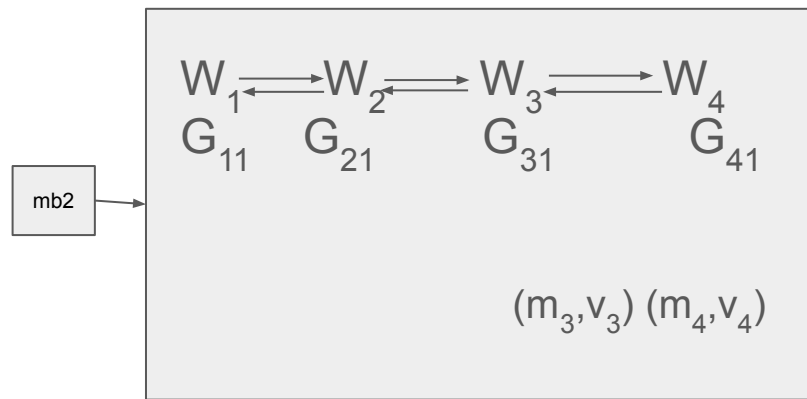
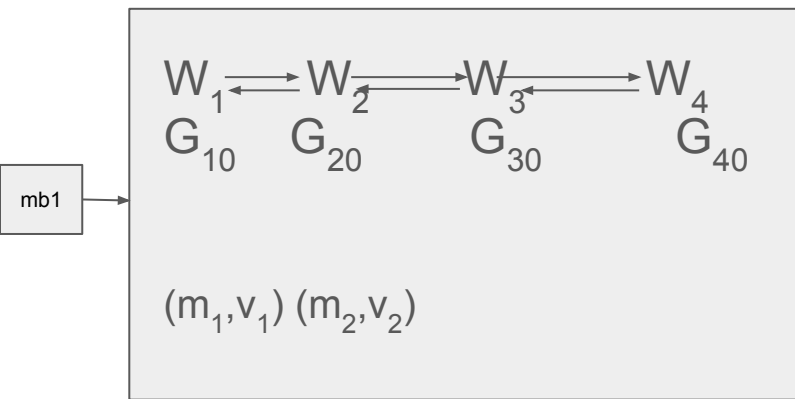


- Update W_1, W_2, W_3 and W_4 on device 0
- Update W_1, W_2, W_3 and W_4 on device 1
- Update each parameter with the optimizer states
- Optimizations: Bucket gradients and computation-communication overlap

Zero-1

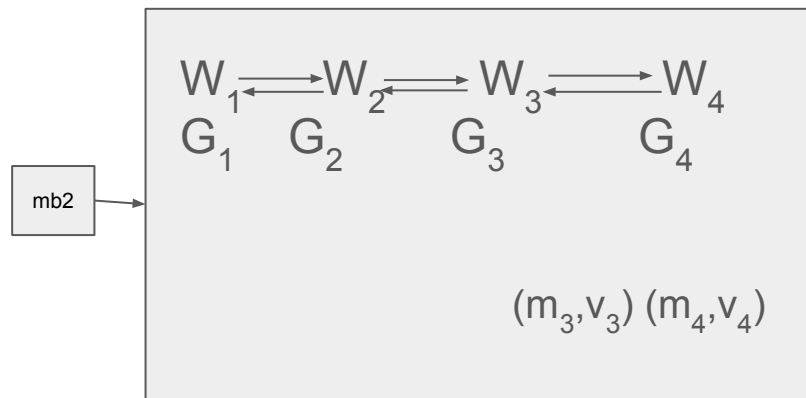
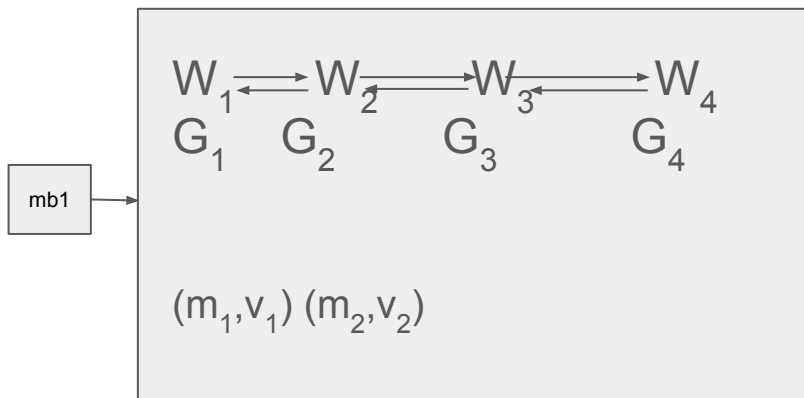


- Shard the optimizers



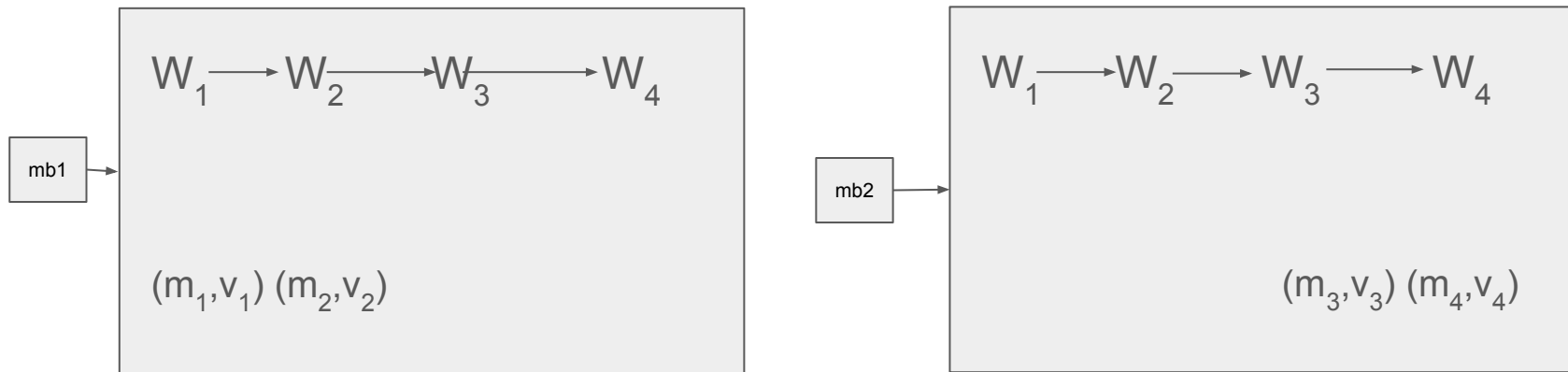
Communication (world_size = 2)

- All-reduce on the gradients
 - $(G_{10} + G_{11}) / 2 = G_1$

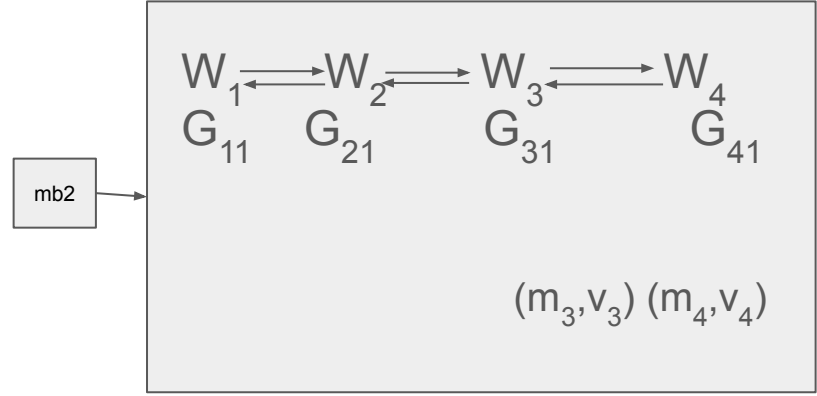
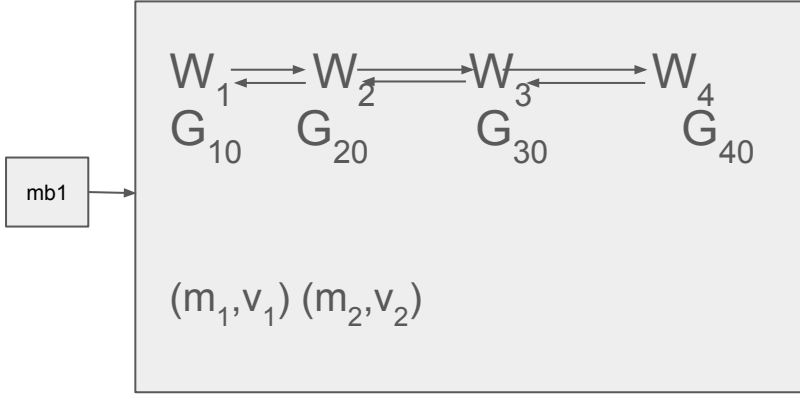


- Update W_1 and W_2 on device 0
- Update W_3 and W_4 on device 1
- All-gather on the weights
 - Send W_1 and W_2 from device 0 to 1
 - Send W_3 and W_4 from device 1 to 0
- Redundant gradients

Zero-2

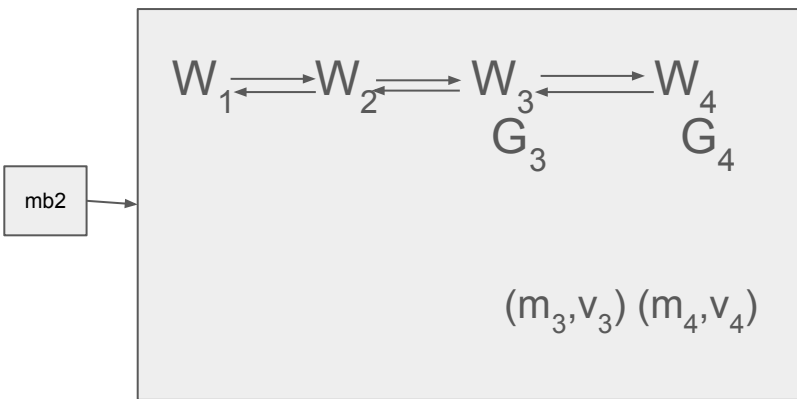
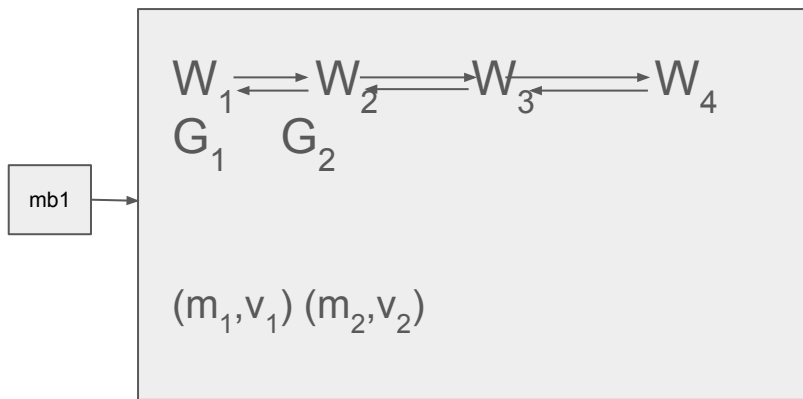


- Shard the optimizers and gradients



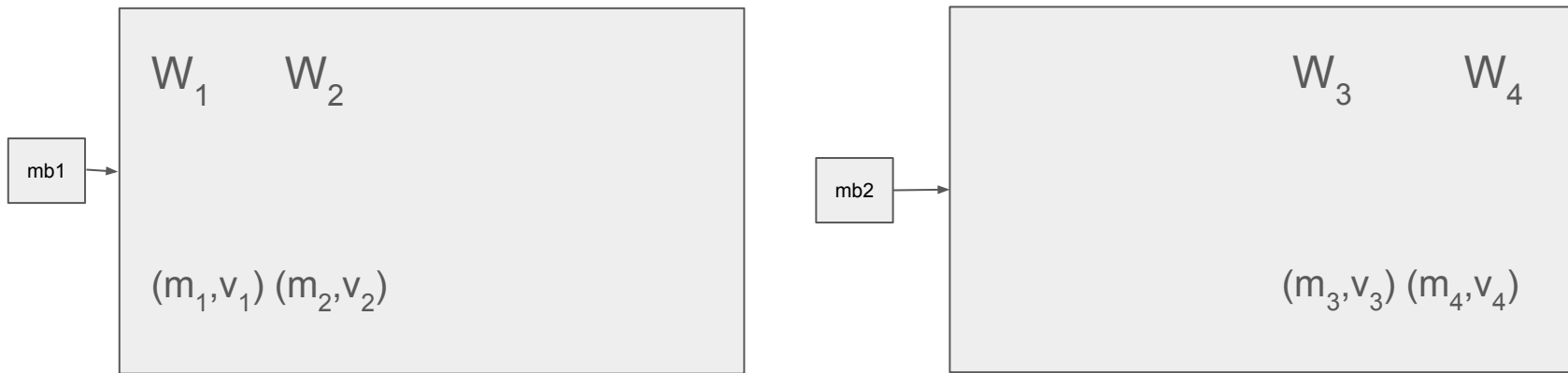
Communication (`world_size = 2`)

- Reduce Scatter on the gradients
 - $(G_{10} + G_{11}) / 2 = G_1$

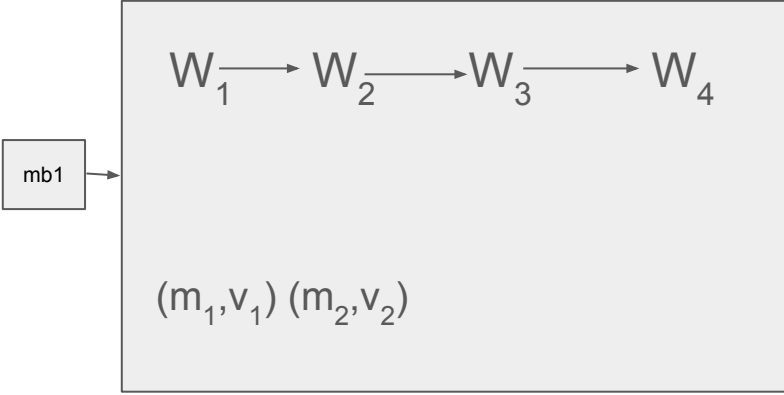


- Update W_1 and W_2 on device 0
- Update W_3 and W_4 on device 1
- All-gather on the weights
 - Send W_1 and W_2 from device 0 to 1
 - Send W_3 and W_4 from device 1 to 0
- No more redundant gradients and reduce scatter has less overhead than all-reduce

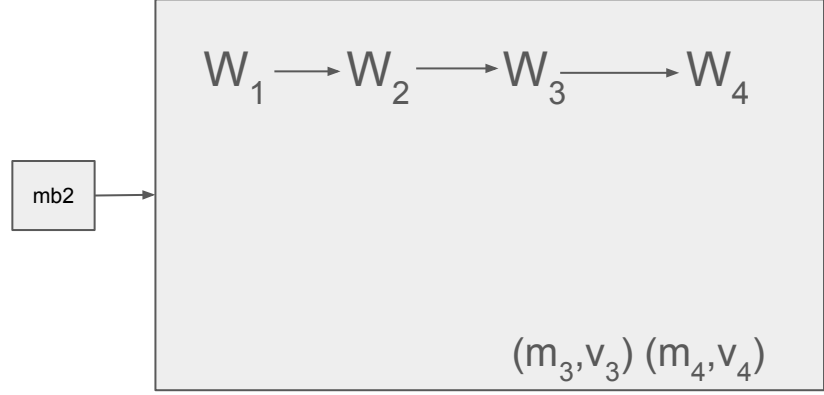
Zero-3

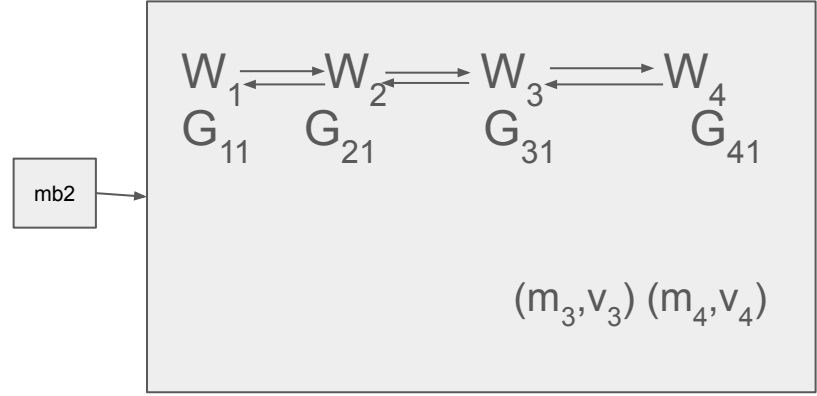
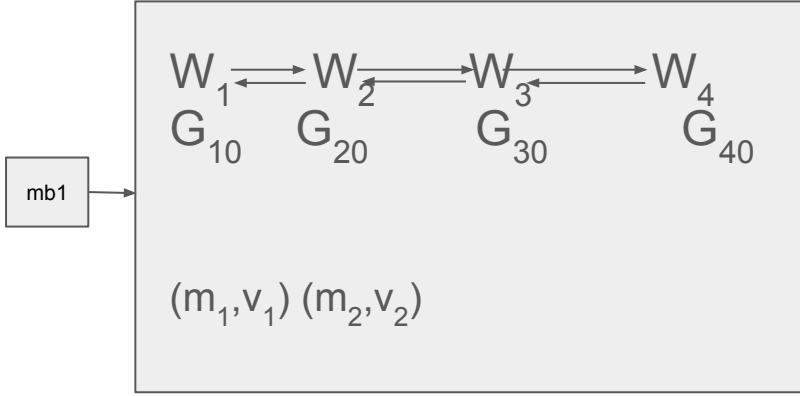


- Shard the optimizers and gradients and weights



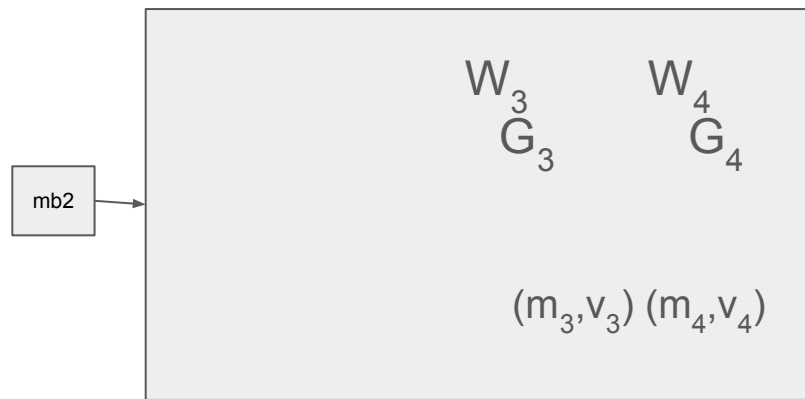
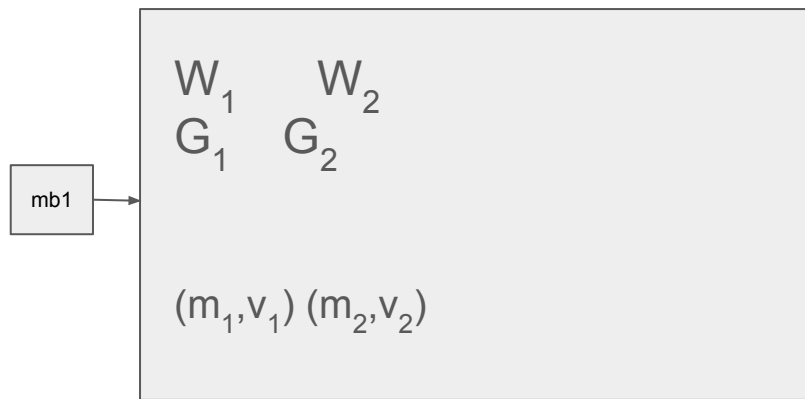
- All-gather on the weights





Communication (`world_size = 2`)

- Reduce Scatter on the gradients
 - $(G_{10} + G_{11}) / 2 = G_1$



- Update W_1 and W_2 on device 0
- Update W_3 and W_4 on device 1
- NO MORE ALL-GATHER